



451/1

COMPUTER STUDIES

PAPER 1

(Theory)

TIME: 2 ½ HOURS

MARKING SCHEME*Kenya Certificate of Secondary Education (KCSE)*

1. a) Define a system unit. (1 mk)
Is the part which houses the brain of the computer.
- b) Two functions of the control unit. (2 mks)
 - i) Coordinate and controls all processing activities in the CPU.
 - ii) Determine which operation or instruction to be executed.
2. a) Define a computer laboratories. (1 mk)

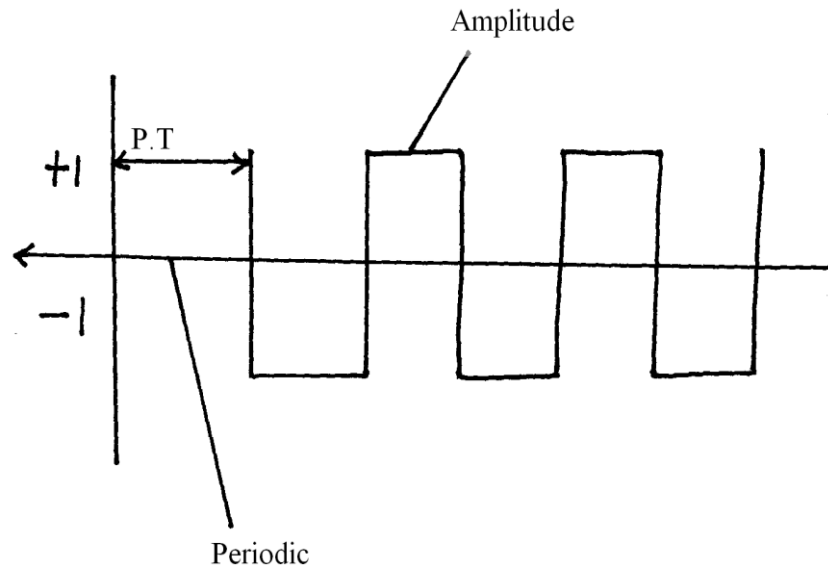
It is a special room set a side and prepared specifically for safe installation and use of computers.
- b) State two functions of the ups. (2 mks)
 - i) Provides power temporarily in the event of a blast
 - ii) It regulates power from an unstable source.
 - iii) Allow the user to switch off the computer correctly and save the work.
 - iv) Safeguards the computer against damage.
3. Discuss the following types of computers. (2 mks)
 - i) Dedicate purpose computers

- Are computers designed to perform a variety of tasks when given the software, but they have been set a side to perform a specific task.
 - ii) Embedded computers

- Are computers that are incorporated or added onto other devices such as phones
4. State two advantages of using biometric devices in voting. (2 mks)
 - 1) Because it is able to capture the voters' attributes ie finger prints and facial recognition



- 2) One cannot vote using other person's voting card
5. Discuss one are computers can be used in science and research. (2 mks)
- i) Weather forecasting – making weather predictions more accurate and reliable.
 - ii) Medical research – For diagnosis, keeping patients' records, inventory control.
 - iii) Military and space exploration science – Computers are used to make research, design, development and control of unmanned spaceships, aeroplanes and missiles.
6. State two types of database models. (2 mks)
- i) Flat file
 - ii) Relational
 - iii) Network
 - iv) Hierchical (award a mark for any 2)
7. a) Differentiate between an Orphan and a widow with reference to word processing. (2 mks)
- An Orphan is the first line of a paragraph that appears as the last line of a page while a widow is the last line of a paragraph tat appears of the first line of a page.
- b) By use of examples, differentiate between a function and a formula in spread sheets. (2 mks)
- Functions are in built predicted formulae that the user can quickly used instead of having to create a new each time eg = Sum(B1 + B4)
- A formula is a mathematical expressions that create a relationship between cells to return a ne value eg = D3 + D4
8. With the aid of a diagram, explain amplitute and periodic time. (3 mks)



Amplitude – Is the maximum displacement that a waveform of an electrical signal can attain.

Periodic time – Time taken by a signal to complete a cycle.

9. a) Explain industrial espionage. (1 mk)

It is spying on a competitor to get information that can be used to cripple the competitor.

- b) Describe data encryption. (2 mks)

-It is done by mixing up into a form that only the sender and the receipt is able to understand.

- The message to be encrypted is called plan text document.

- After encryption an algorithm/ key is sent as a cipher text on the network.

10. What are data terminal equipment? (1 mk)

- These are devices used to process host and terminal data on a network eg mobile phones, PDAS, computers.

11. a) Explain open learning. (1 mk)

- This is where learning materials and lectures are made available online over the internet.

- b) Computers are used to enhance marketing in a variety of ways, one of which is E-business. Explain how computers are used in E-business. (2 mks)



- i) Advertisement
- ii) Buying of goods an affordable price
- iii) Variety of goods.

12. Discuss two computational errors giving example in each case. (4 mks)

i) Overflow errors

- It occurs if the results from a calculation is too large to be stored in the allocated memory space eg If a byte represented using 8 bits, an overflow will occur if the results of the calculation gives a bit (1)00111011

ii) Truncation error

- When a real number have a long fractional part which is cut off eg
0.784968 -0.984

iii) Rounding errors

- Is as a result of raising or lowering a digit in a real numbers to the required rounded number eg 30.666 – 30.7

13. Differentiate between worm and a virus. (2 mks)

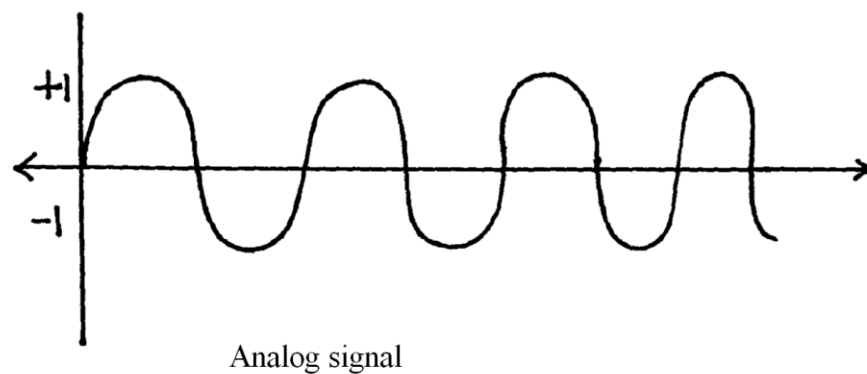
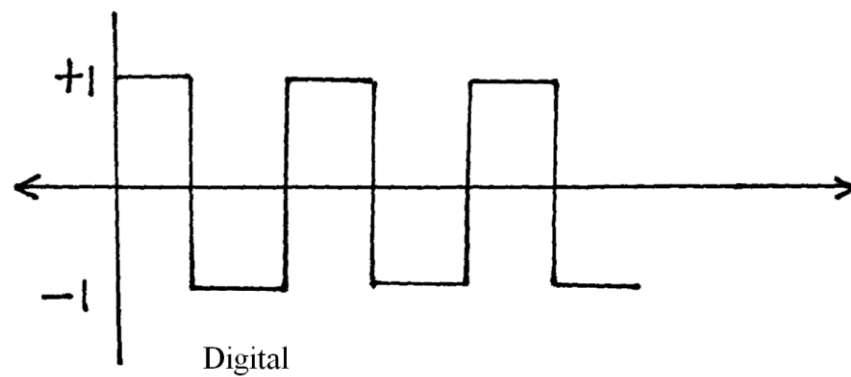
- A worm is a malicious program that self-replicates hence dogs the system memory and storage media a virus is a destructive program that attaches itself on a removable drive and causes damage to computer systems such as deleting system files.

14. Differentiate sequential file organization from indexed sequential file organization. (2 mks)

- Sequential file organisation records are stored and accessed in a particular order sorted using a key field while indexed sequential records are stored and accessed in a particular order using an index to locate individual records.

15. With the aid of a diagram, differentiate between analog and digital data signals.

Digital signals are discrete in nature and are represented using two states ie ON and OFF while analog signals are continuous in nature.



SECTION B

Answer question 16 and any other three questions in this section.

16. a) State two types of selection controls used in high level programming language. (2 mks)
- i) IF---THEN
 - ii) IF---- THEN----ELSE
 - iii) NESTED IF selected
- b) Give two characteristics of a good program. (2 mks)
- i) Key words should be properly used
 - ii) Should execute and produce the required results
 - iii) Should have a start and end



c) Use the flow chart below to answer the questions that follow:

i) Write a pseudocode for the above flow chart. (6 mks)

Start

Repeat

Input C, X and Y

Print "Error"

Until $C > 0$

Set $Z = X * Y$

Print Z

Set $C = C - 1$

iii) Modify the flow chart so that the program does not accept any negative inputs. (3 m)

17. a) State and explain two disadvantages that will come about if a network was to be installed in your school. (4 mks)

i) High initial cost – The cost of buying network hardware and software is very high.

ii) Security issues – The data on the network is prone to illegal access threats.

b) Disadvantages of wireless networks. (4 mks)

i) Difficult to establish or configure

ii) Initial cost is very high

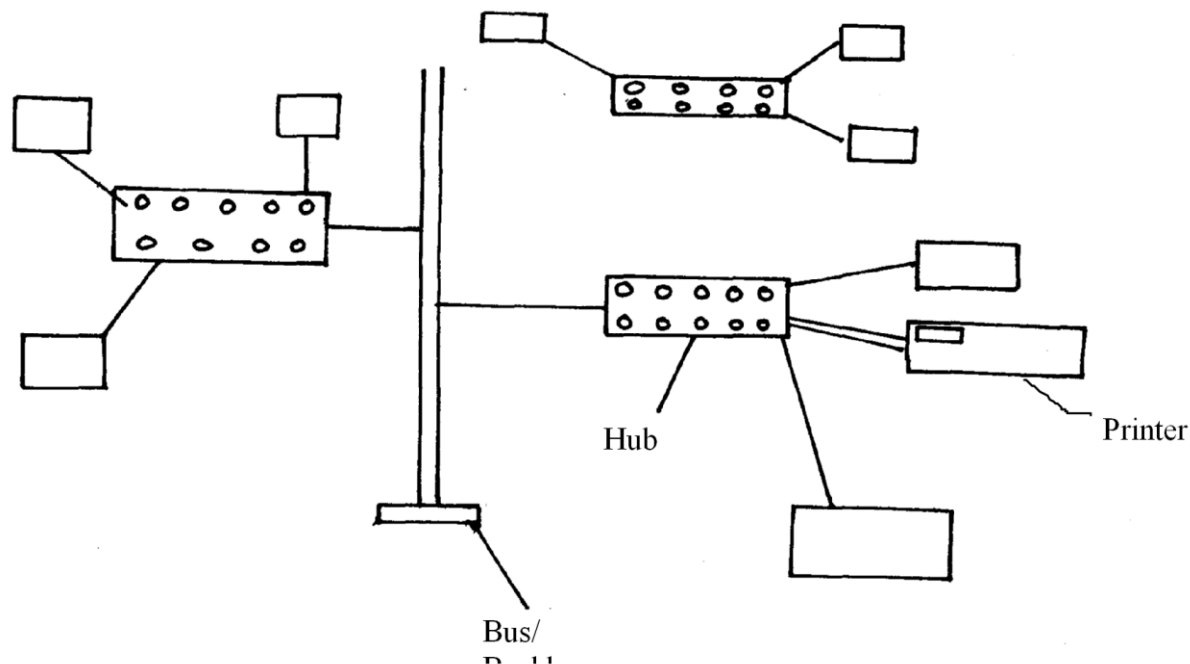
c) Write the following abbreviations in full. (4 mks)

i) F.T.P – File transfer protocol

ii) H.T.T.P – Hyper text transfer protocol

d) With the aid of a diagram, discuss Hybrid topology. (3 mks)

- This are groups of star – Configured networks connected to a linear bus backbone.



- e) Discuss one advantage of a client/ server network. (2 mks)
- It is easier to access files and other resources easily because they are all available on the server computer.
18. a) Explain why a computer is able to display the correct time and date when it has just be switched on. (2 mks)
- Because it is set and stored in a special memory known as the complementary metal-oxide semiconductor (cmes) which is a powered cell.
- b) Discuss two types of special memories found I computer system. (4 mks)
- i) Cache – Fast type of RAM found in the processor.
 - ii) Buffers – Special memories found in the input/ output devices.
 - iii) Register – Hold one piece of date at a time and are found in the ALU
- c) i) Define a Bus with reference to a computer system. (1 mk)
- Bus – This are electronic pathways or links
- ii) List to examples of buses. (2 mks)
- i) Address bus



- ii) Data bus
 - iii) Control bus
- d) Distinguish between a power cable and interface cable. (2 mks)
 - A power cable is used to transmit data from the source (socket) to the computer while interface cables are special cables used to connect peripheral cables to the system unit.
- e) Differentiate between the different types of RAM. (4 mks)
 - Static RAM – is a fast type of RAM located inside the microprocessor
 - Dynamic RAM – is a relatively slow type of RAM. It has the tendency for the stored charge to leak a way even with constant power supply hence requires periodic recharging.
- 19. a) i) Define a system. (1 mk)
 - Is a set of organised components which intersect in a given environment and within a specified banday to achieve collective goals.
- ii) Explain system entropy. (1 mk)
 - It means decay where a system slowly becomes useless to the user due to improvement in user requirements.
- b) State three circumstances that can lead to development of information systems. (3 mks)
 - i) New opportunity
 - ii) Problems
 - iii) Directives
- c) Distinguish parallel changes over from straight change over as used in system implementation. (2 mks)
 - Parallel changeover both old and new systems are run parallel to end other for sometime until users have confidence in the new system is stopped and discarded and the new system started immediately.
- d) Discuss two fact finding methods. (4 mks)
 - i) Interviews – It is carried out with relevant stakeholders inorder to get views a bout the current system.



- ii) Questionnaires – Special purpose document that allows a person to collect information and opinions from people who receive and respond to it.
 - iii) Observation – Require the observer to participate or watch closely as a person performs activities in order to learn about the system.
 - iv) Automated method – Used when one wants to get actual data which may be difficult to get through interviews, observation or questionnaires.
- e) Differentiate an open system from a closed system. (2 mks)
- f) List two responsibilities of a system analyst. (2 mks)
- i) Review an existing system and making recommendation on how to improve or implement an alternative system
 - ii) Working hand in hand with programmers to construct a computerised system.
 - iii) Coordinating training of new system users and owners.
20. a) State two ways of representing a signed number. (2 mks)
- a) Using ones compliment
 - b) Using twos compliment
 - c) Prefixing an extra sign bit to a binary number.
- b) Write the following abbreviations in full. (2 mks)
- i) ASCII – American standard code for information interchange
 - ii) EBCDIC – Extended binary coded decimal interchange code
- c) Convert 110011.0110 to Decimal (3 mks)

P.V	2^4	2^3	2^2	2^1	2^0	.	2^{-1}	2^{-2}	2^{-3}	2^{-4}
B.D	1	1	0	1	1	.	0	1	1	0
D.V										



$$= 8 + 6 + 0 + 2 + 1 \cdot 0 + 0.25 + 0.125 + 0$$

$$= 17.000 + 0.375$$

$$= 17.375_{10}$$

d) Differentiate between absolute value and base value. (2 mks)

i) It is easier to construct machines (electrical circuits) based on binary or ON and OFF.

ii) Digital devices are more reliable, small and use less energy

e) Outline two reasons for using binary system in computers. (2 mks)

f) Perform $111011001000_2 - 11011011001_2$ using ones complement convert your answer to Hexadecimal. (4 mks)

$$= 111011001000 - 11011011001$$

$$= 111011001 + (-11011011001)$$

$$\text{ones complement} = 00100100110$$

$$= 111011001000$$

$$\underline{00100100110}$$

$$111111101110$$

$$= \underline{1111} \quad \underline{1110} \quad \underline{1110}$$

$$\text{F} \quad \text{E} \quad \text{E}$$

$$= \text{FEE}_{16}$$